

47. We learned in [Section 14-6](#) that electrophilic additions to conjugated dienes at low reaction temperatures give kinetic product ratios. Furthermore, these kinetic mixtures may change to mixtures with thermodynamic product ratios when the temperature is raised. Do you think that cooling the thermodynamic product mixture back to the original low reaction temperature will change it back to the original kinetic ratio? Why or why not?

48. Compare the addition of H^+ to 1,3-pentadiene and 1,4-pentadiene (see [Problem 46](#)). Draw the structures of the products. Draw a qualitative reaction profile showing both dienes and both proton addition products on the same graph. Which diene adds the proton faster? Which one gives the more stable product?

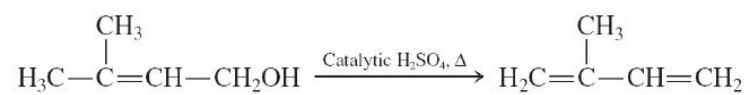
49. What products would you expect from the electrophilic addition of each of the following reagents to 1,3-cycloheptadiene? **(a)** HI; **(b)** Br₂, Br₂, in H₂O; H₂O; **(c)** I₂, I₂; **(d)** H₂SO₄, H₂SO₄ in CH₃CH₂OH. CH₃CH₂OH. (Hint: For b and c, see [Exercise 14-13](#), specifically drawings A-C.)

50. Give the products of the reaction of *trans*-1,3-pentadiene with each of the reagents in [Problem 49](#).

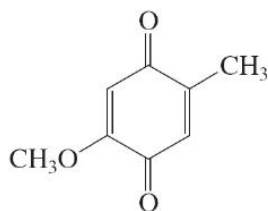
51. What are the products of reaction of 2-methyl-1,3-pentadiene with each of the reagents in [Problem 49](#)?

53. Give the products expected from reaction of deuterium iodide (DI) with **(a)** 1,3-cycloheptadiene; **(b)** *trans*-1,3-pentadiene; **(c)** 2-methyl-1,3-pentadiene. In what way does the observable result of reaction of DI differ from that of reaction of HI with these same substrates [compare with parts (a) of [Problems 49](#) through [51](#)]?

56. Dienes may be prepared by elimination reactions of substituted allylic compounds. For example,



76. Diels-Alder cycloaddition of 1,3-butadiene with the cyclic dienophile shown below takes place at only one of the two carbon-carbon double bonds in the latter to give a single product. Give its structure and explain your answer. Watch stereochemistry.



This transformation was the initial step in the total synthesis of cholesterol ([Section 4-7](#)), completed by R. B. Woodward (see [Section 14-9](#)) in 1951. This achievement, monumental for its time, revolutionized synthetic organic chemistry.

Team Problems